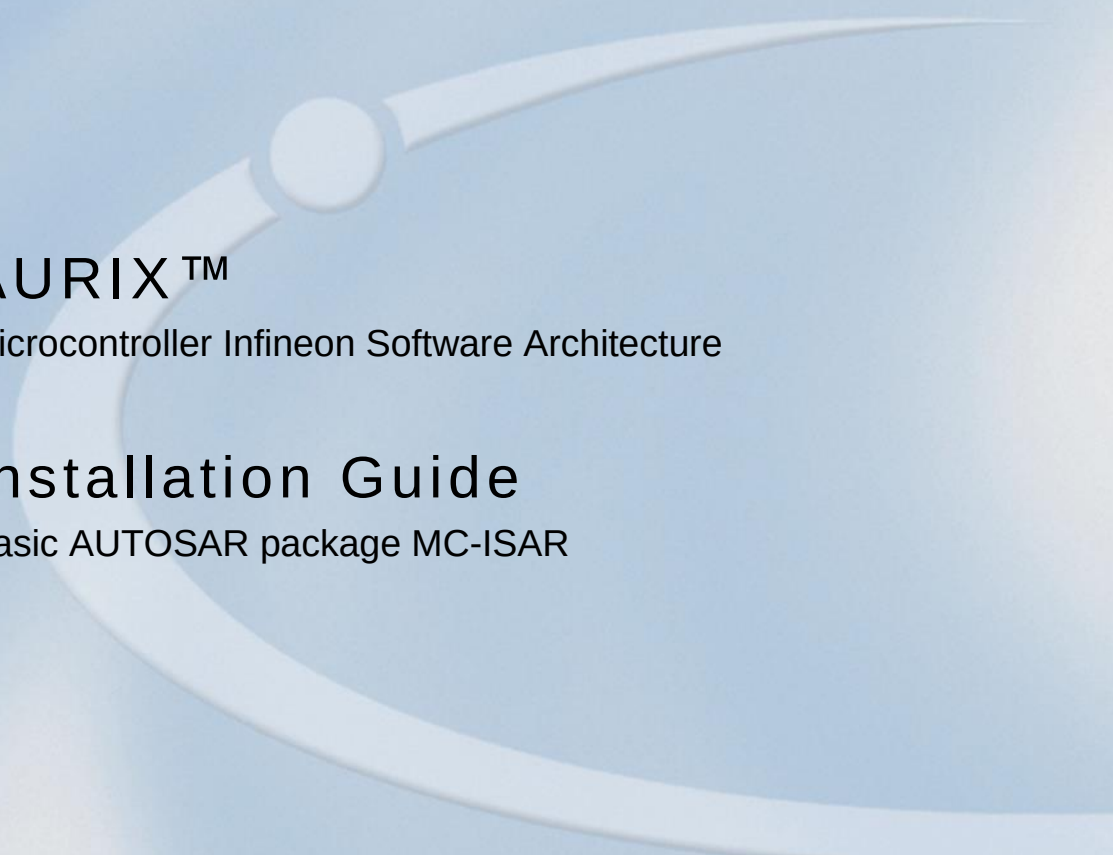


A large, light blue, stylized graphic of a microcontroller package, possibly a BGA, is centered on the page. It features a circular base with a small circle in the center, and a curved line extending from the top left, ending in a small circle, resembling a pin or a connection point.

AURIX™

Microcontroller Infineon Software Architecture

Installation Guide

Basic AUTOSAR package MC-ISAR

Release 1.5

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Date	Version	Change Description
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2011-11-09	0.2	Changed for Aurix W6 Alpha release.
2012-05-16	0.3	Updated the figures.
2012-10-22	0.4	Added changes for 3XX and 4XX alpha release
2013-01-03	0.5	Updated for V200 release.
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2015-05-19	1.0	Updated as per review comments and for changes in the Packages
2015-07-10	1.1	Added chapter 2.3.2 Sub-Derivative Selection.
2015-10-13	1.2	Added chapter 2.3 update for Tresos Eclipse issue
2016-06-15	1.3	Moved the chapter Sub-Derivative Selection to section 2.3.2.1. In section 2.3.2, added information that sub derivative needs to be selected first before adding modules.
2016-06-15	1.4	Corrected based on review comments.
2016-06-18	1.5	Updated for CAN Enhanced Driver DemoApp Configuration in section 2.3.2

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1 Introduction

This document explains on the installation and Tresos configuration in order to run a demo application for AURIX_MC-ISAR project.

1.1 Purpose of the document

This document describes the delivery folder structure of AURIX_MC-ISAR to its users. Additionally it helps the user to run the setup and to configure tresos for running the demo application by providing demo diagrams.

1.2 Scope

The scope is limited to explaining the installation and configuration of setup only. Port pin connections to run the demoapp are explained in Demo_App.txt part of BASE package. Tresos configuration is explained only for AUTOSAR 4.0.3 environment. For the actual version of the tresos tool to be used, refer release notes.

Installation procedure and Tresos Configuration concept holds good for all AURIX Products.

For example: A base package name will have the following convention

MC-ISAR_AS<xx>_AURIX_TC<Con>_<Step>_PB_BASE_V<nnn>

Here <xxx>: 4XX / 3XX

<Con> : 27X / 26X / 29X / 23X / 22X / 21X

<Step> : AB / BA / BB / CA / DB

<nnn>: package version number :200

As per the given package, suitably these variables can be changed and the procedure to install remains the same.

1.3 Definitions

Table 1 Definitions

Keyword	Definition
BASE	A package consists of MCU, PORT, DIO, ADC, SPI, GPT, ICU, PWM, GTM and WDG driver. This also has the support modules like ECUM and IRQ
COM_BASIC	A package consists of LIN, CAN and CanTrcv driver
COM_ENHANCED	A package consists of Flexray and Ethernet driver
MEM	A package consists of FLS and FEE driver
COMPLEX / CD	A package consists of FlsLoader, MSC, UART and DMA driver
DEMOCOD	A package consists of SENT, HSSL, STM, DSADC, SMU, IOM and I2C driver
STDLINCD	A package consists of StdLin driver
LIB	A package consists of CRC and BFX library

1.4 Abbreviations

The abbreviations used inside this document are explained in [Table 2](#)

Table 2 Abbreviations

Abbreviation	Explanation
AUTOSAR	AUTomotive Open System Architecture
BSW	Basic SoftWare bundle
DEM	Diagnostic Event Manager
DET	Development Error Tracer
ELF	Executable & Linkable Format
GPT	General Purpose Timer
MCAL	MicroController Abstraction Layer
MCU	MicroController Unit
OS	Operation System
SW	Software
SSC_ROOT	Standard Software Core root directory
BMD	BSW Module Description
SPI	Serial peripheral Interface

2 Installation

This section describes a use case to install the BASE and MEM packages and configure the same in Tresos.

Note: All the Screen shots are taken from Aurix TC29x.

2.1 BASE Package

Run MC-ISAR_AS<xxx>_AURIX_TC<con>_<Step>_PB_BASE_V<nnn>.exe.

For TC29x Release: Run MC-ISAR_AS<xxx>_AURIX_TC29X_<Step>_PB_BASE_V<nnn>.exe.

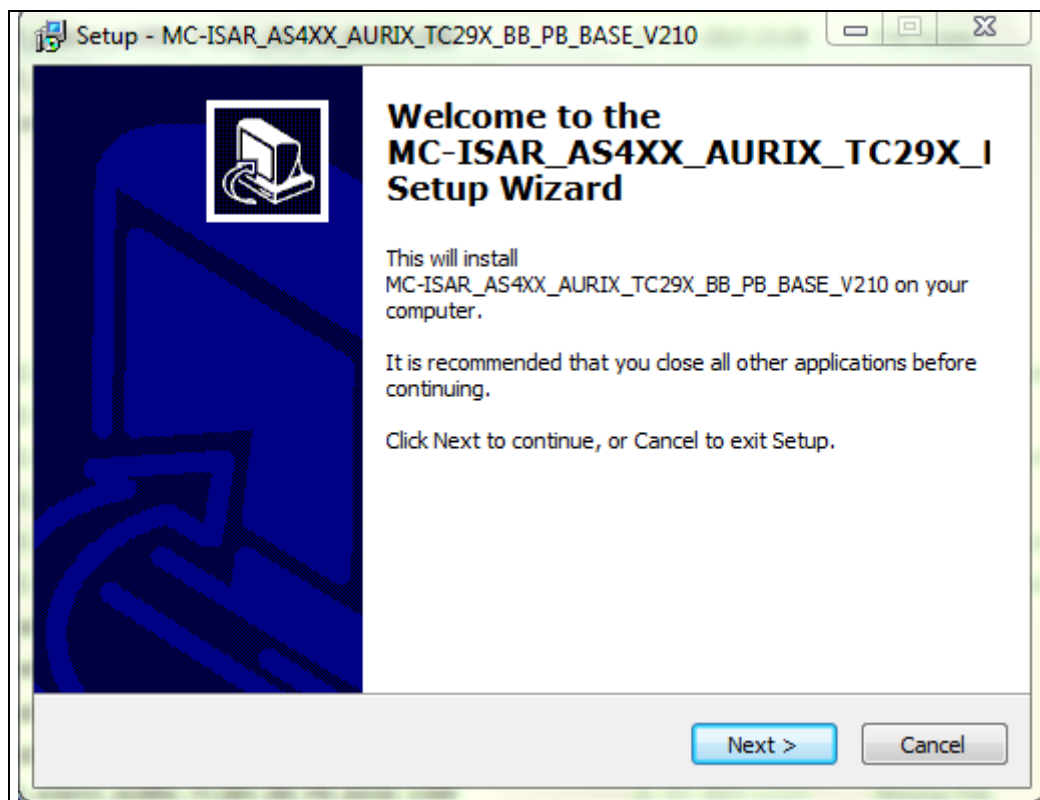


Figure 1 BASE package installation window

Click **Next** in [Figure 1](#)

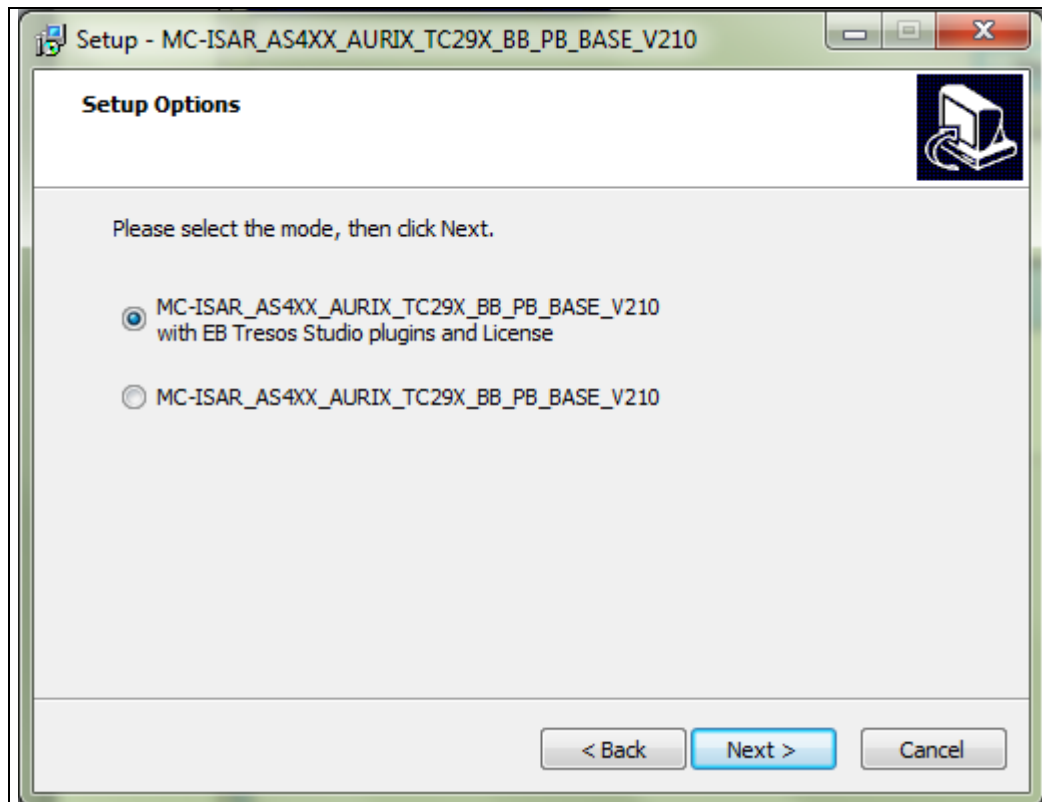


Figure 2 Base Package : mode selection

Select the required mode and click **Next** in [Figure 2](#)

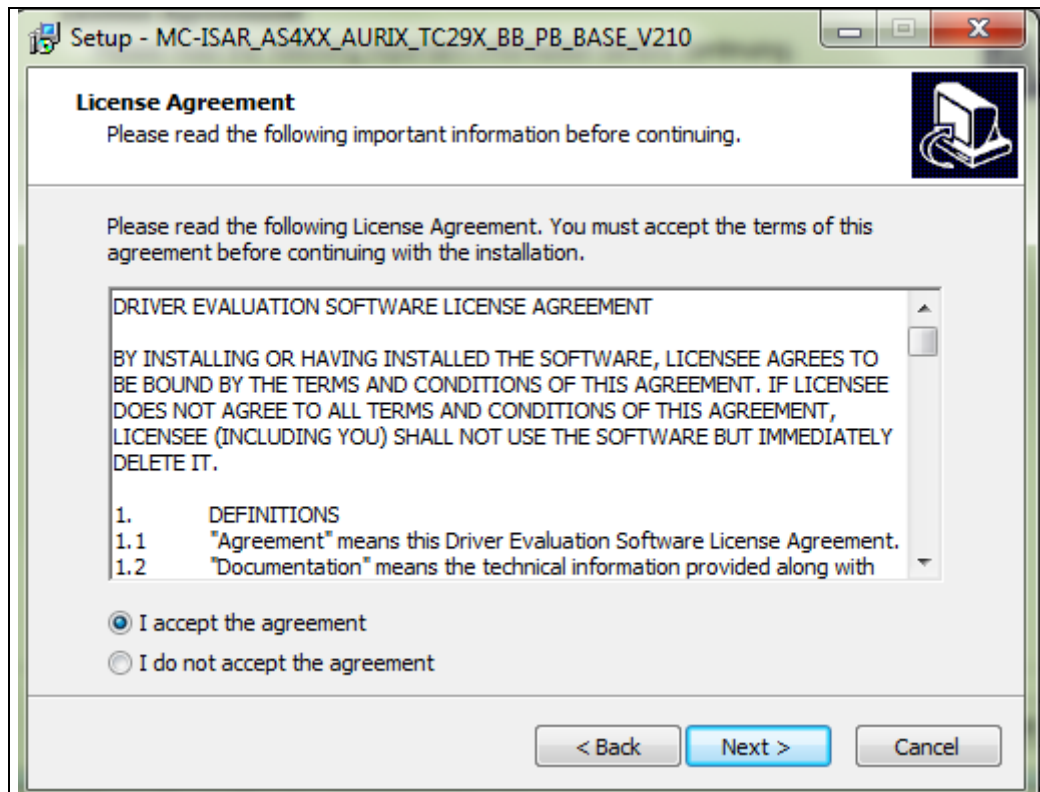


Figure 3 Base Package : License Agreement

Accept the agreement and click **Next**

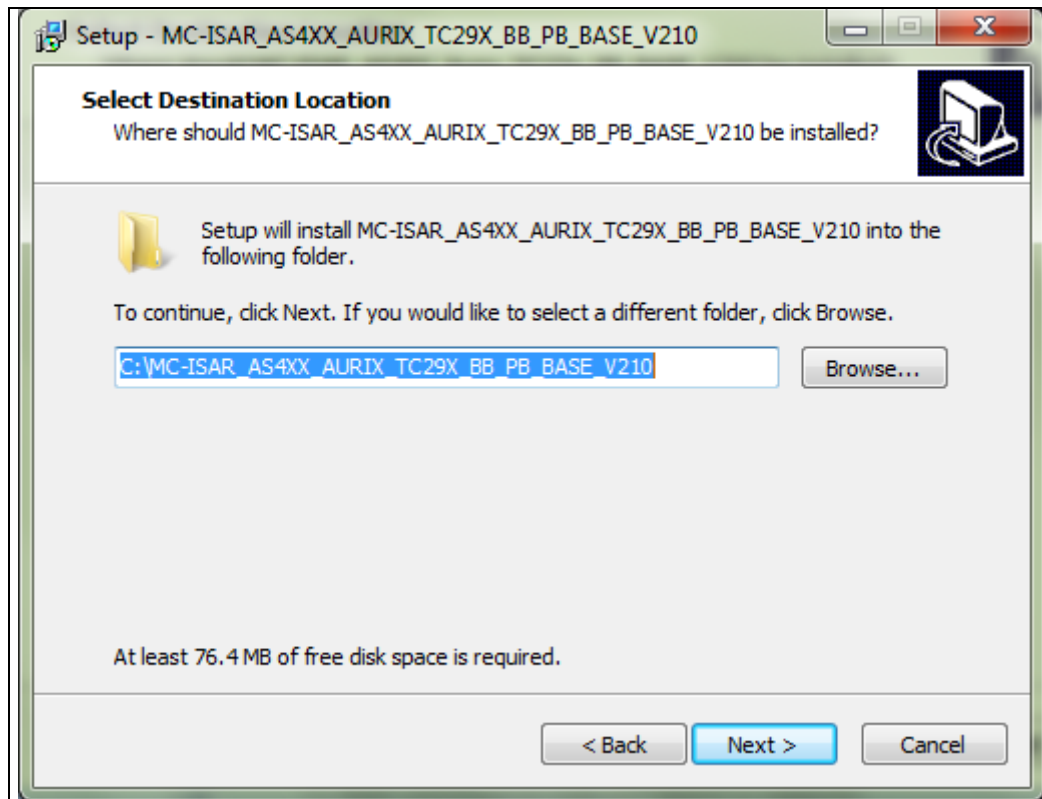


Figure 4 Base Package : Window to set destination folder

Figure 5 shows window to select a folder to install EB Tresos plugins. This window appears only if the first mode is selected in **Figure 2**. If your tresos installation is done in the folder C:\EB\tresos\2012a then the path that should be given in the edit box should be C:\EB\tresos\2012a\plugins.

Note: If the plugins already exists in above path, they will be overwritten.

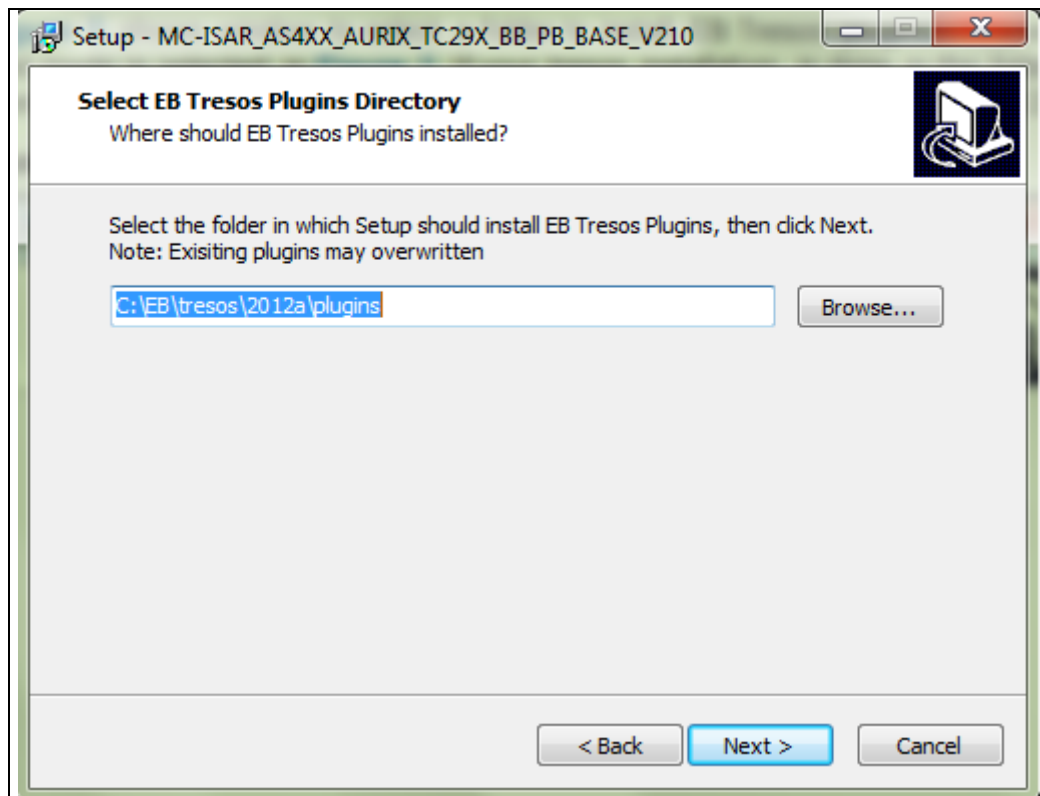


Figure 5 Base Package: Dialog to accept plugins path

After clicking **Next** similarly point to the configuration folder where the license files will be copied. With reference to above example the folder name would be C:\EB\tresos\2012a\configuration.

Note: If license already exists, it will be overwritten.

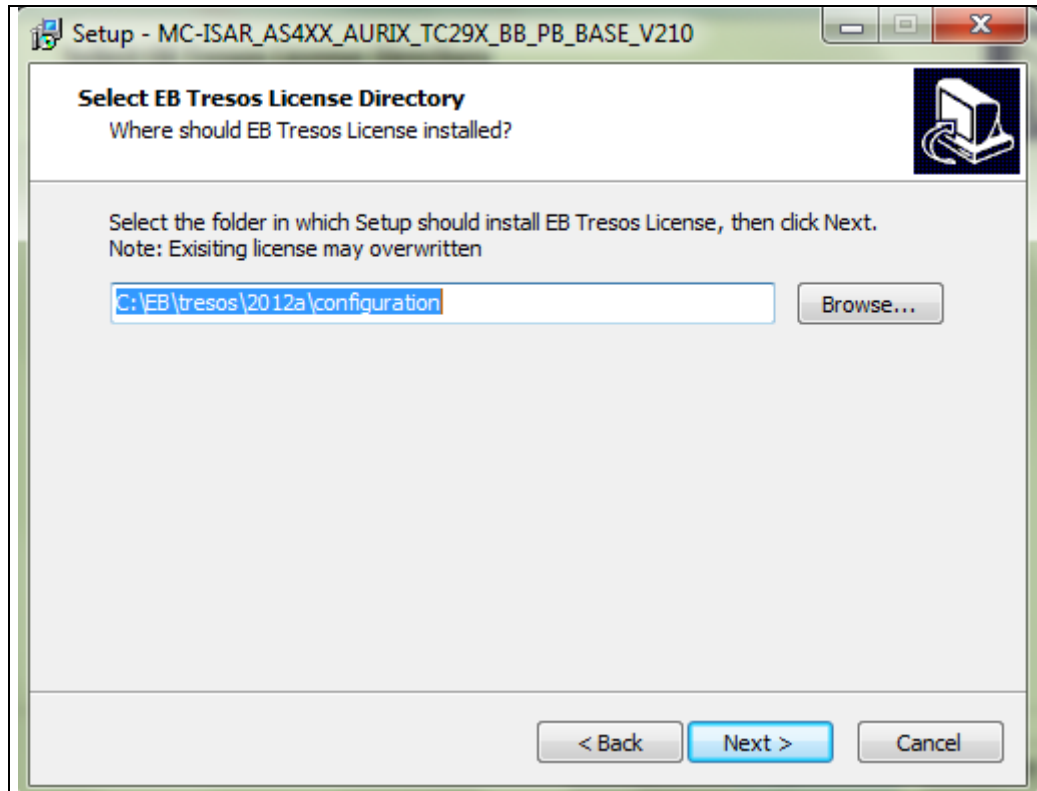


Figure 6 Dialog to accept license path

After this step, BASE package will get installed on the required destination folder. Folder directory after installation is shown in [Figure 7](#)

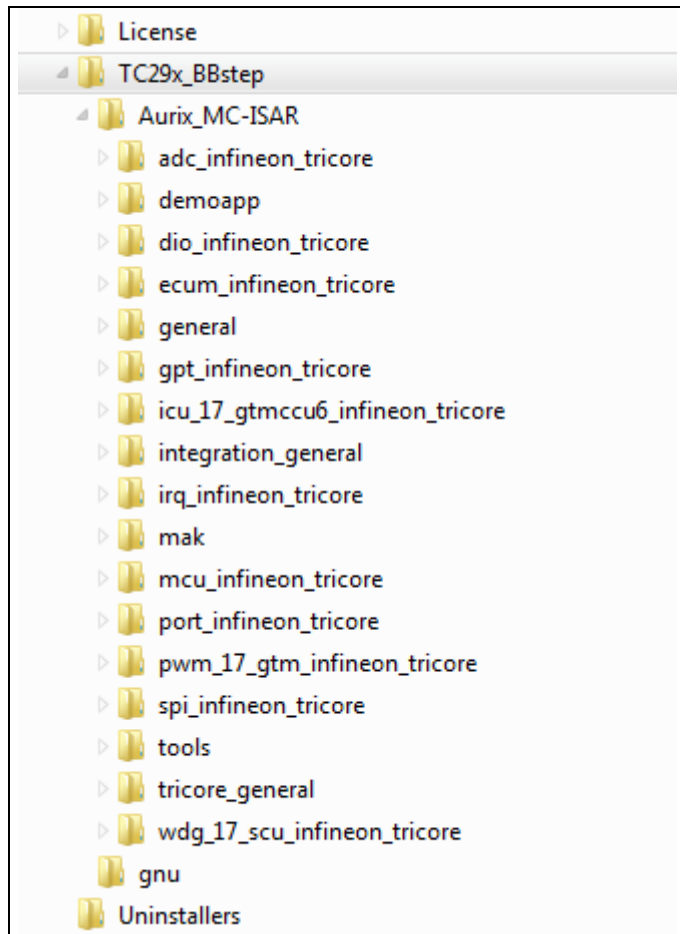


Figure 7 BASE: folder structure after installing BASE

After BASE package installation GPT driver folder structure is shown in [Figure 8](#)

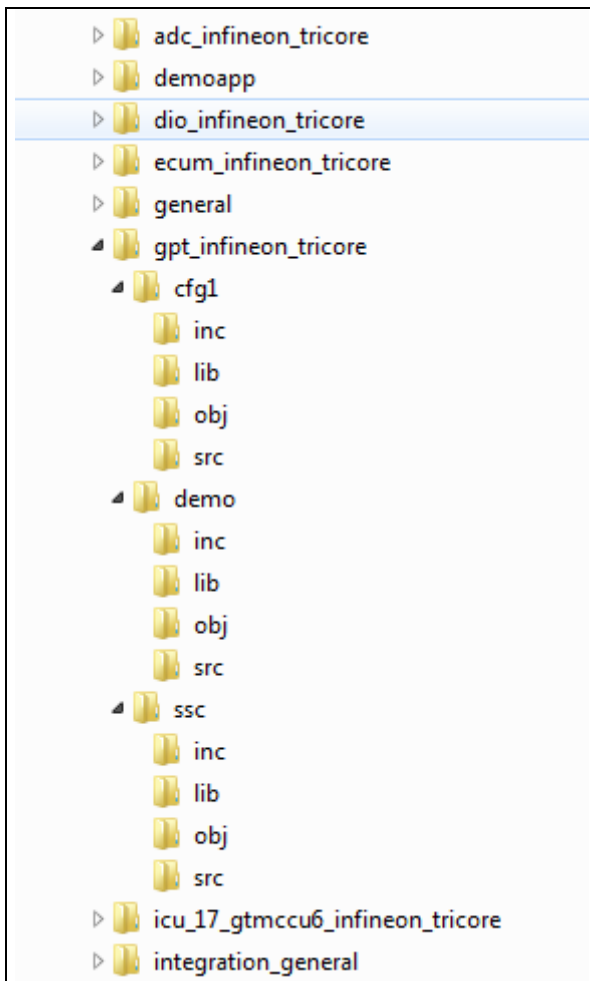


Figure 8 Folder structure with GPT

Each <Module>_infineon_tricore, will have three sub folder namely ssc, cfg1 and demo as shown in the [Figure 8](#) , where

ssc folder contains driver source files,

cfg1 folder contains configuration files generated from Tresos for demo configuration,

demo folder contains <Module> specific demo application.

After installing the package plugin folder structure for Aurix is shown in [Figure 9](#).

Note: Close Tresos, if already open in your system.

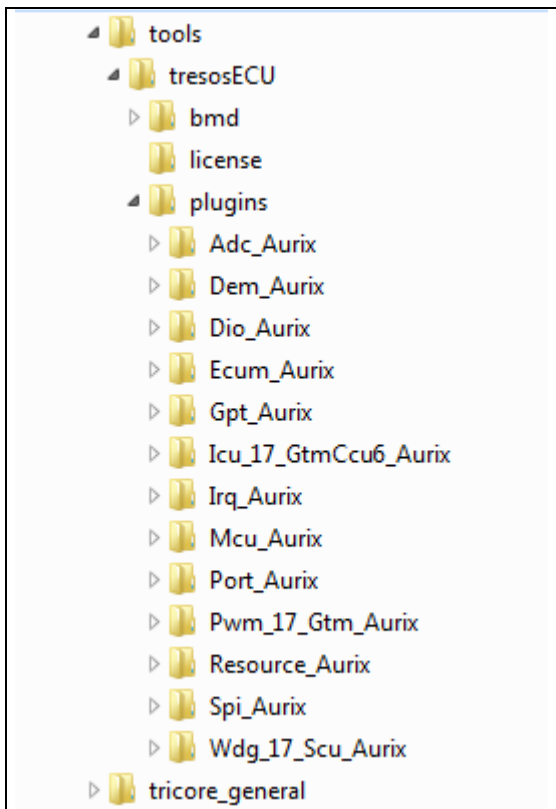


Figure 9 Folder structure of the plugins

2.2 MCAL Packages

Install any MCAL package in the same directory as the BASE package.

To install any MCAL package the following steps are required.

MEM is taken as example here

Similar to BASE package the installer will display similar options. As recommended click the option to copy into Tresos folder as well. Similar to Figure4, Figure 5 gives the required paths.

The folder structure in BASE is shown below after installing MEM (in the same path as the BASE package), you can clearly see the FLS driver folder.

By default installer will install the packages in the default directory (Ex: C:\MC-ISAR_AS4XX_AURIX_TC29X_BB_< Func_Package >_V210), if the user wants to install and run the Demo Application then the MCAL package folder to be copied to the BASE installed directory folder manually. (Ex: C:\MC-ISAR_AS4XX_AURIX_TC29X_BB_PB_< Func_Package>_V210 to MC-ISAR_AS4XX_AURIX_TC29X_BB_PB_BASE_V210).

The above manual step can be avoided, while installing MCAL packages destination directory should be selected as BASE installed directory.

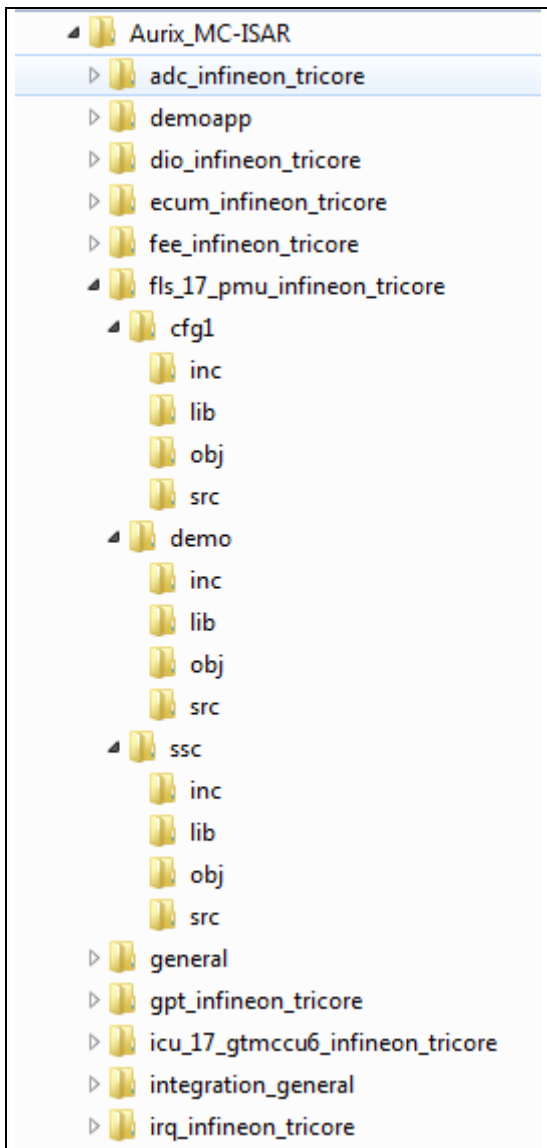


Figure 10 Folder structure with FLS

2.3 Tresos

2.3.1 Tresos - Eclipse Issue

After installing plugins in Tresos from the package installation following should be done:

(Same is also available in 2.4_Studio_documentation_developers_guide.pdf of Tresos 16)

6.3.9. Adding the module to EB tresos Studio

To install the plug-in to EB tresos Studio, copy the whole directory to the EB tresos Studio installation into the directory `plugins`.

Due to a bug in Eclipse, all directories named `org.eclipse.*` underneath the configuration directory of the EB tresos Studio installation have to be removed.

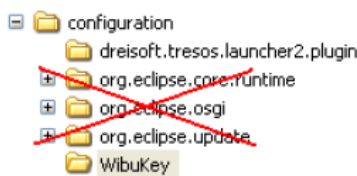


Figure 6.11. Directories to delete

The `org.eclipse.*` directories contain cached information about installed plugins. Not deleting them can cause several undefined effects.

After restarting EB tresos Studio the new plug-in will be available.

If you have purchased EB tresos AutoCore together with EB tresos Studio, you can also integrate the plugin-in with the User Build environment. You will find the instructions on how to do this in the EB tresos AutoCore documentation: [EB tresos AutoCore documentation/EB tresos AutoCore user's guide/Build environment](#).

2.3.2 Tresos Configuration

Follow the below step to configure the CAN Enhanced driver prior to Tresos configuration.

Copy **Can.xdm** file from

C:\MC-ISAR_AS4XX_AURIX_TC29X_BC_PB_CAN_ENHANCED_V100\TC29x_BCstep\Aurix_MC-ISAR\demoapp\cfg\tresos_workspace\Selectable_Cfg to

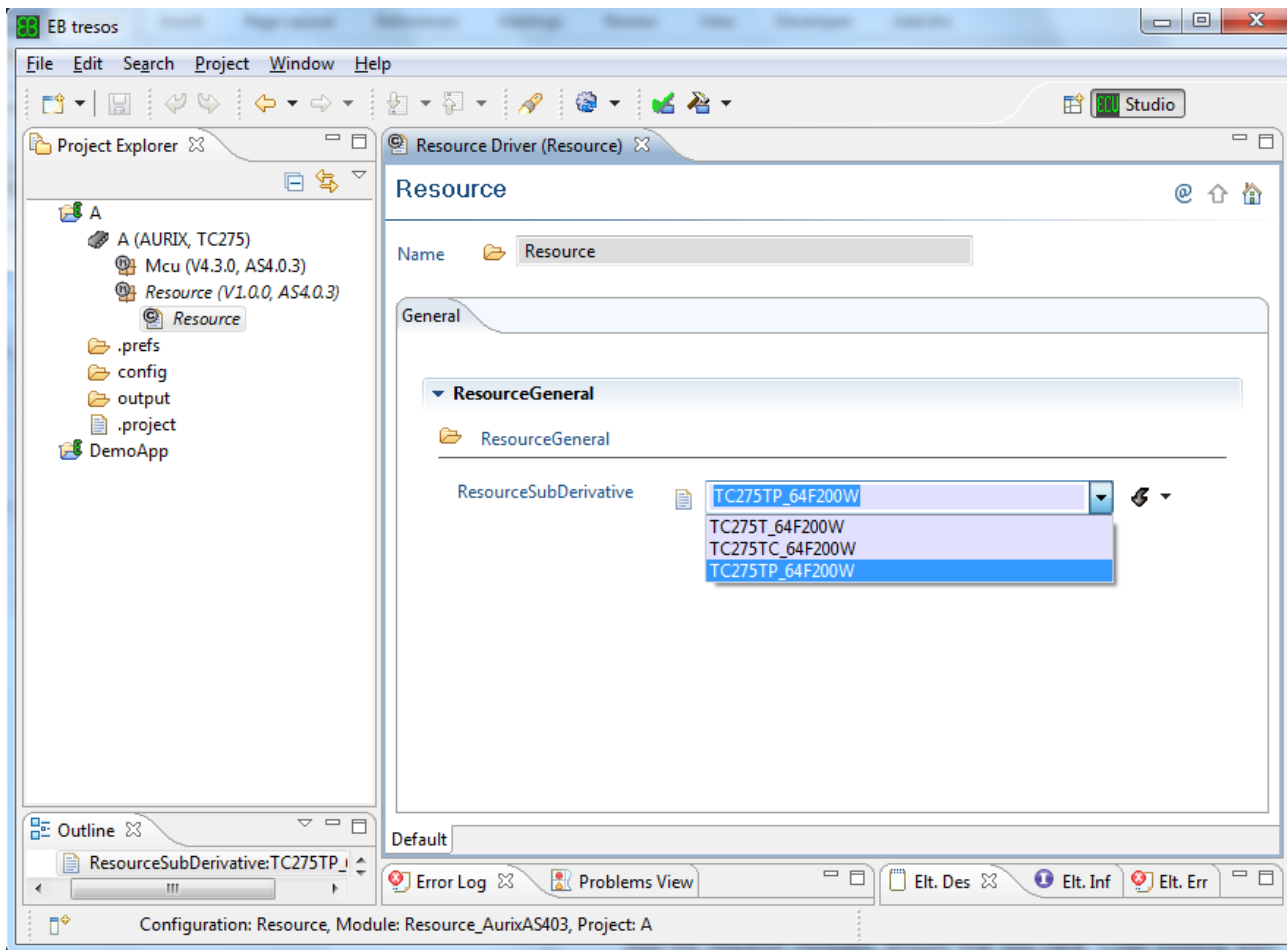
C:\MC-ISAR_AS4XX_AURIX_TC29X_BC_PB_BASE_V320\TC29x_BCstep\Aurix_MC-ISAR\demoapp\cfg\tresos_workspace\Selectable_Cfg\DemoApp\config

Start Tresos and choose File>Switch workspace. Browse to

C:\MC-ISAR_AS<version>_AURIX_TC<Con>_<Step>_PB_BASE_V<package-version>\TC<Con>_<Step>\Aurix_MC-ISAR\demoapp\cfg\tresos_workspace\Selectable_Cfg.

Switch to this workspace, this will take some time as the tresos would close and open again. Select appropriate sub derivative in the resource module and then add the required modules.

2.3.2.1 Sub-Derivative Selection



The ResourceSubDerivative parameter in the Resource module shall be configured for every tresos project. The user shall select the exact sub-derivative from the available list and save the configuration project.

After saving the tresos configuration project, the tresos tool *must* be restarted.

Note: For all new configuration projects ResourceSubDerivative parameter must be configured.

Note: Tresos tool must be mandatorily re-started, each time the value of ResourceSubDerivative parameter is changed.

2.3.2.2 Add a module

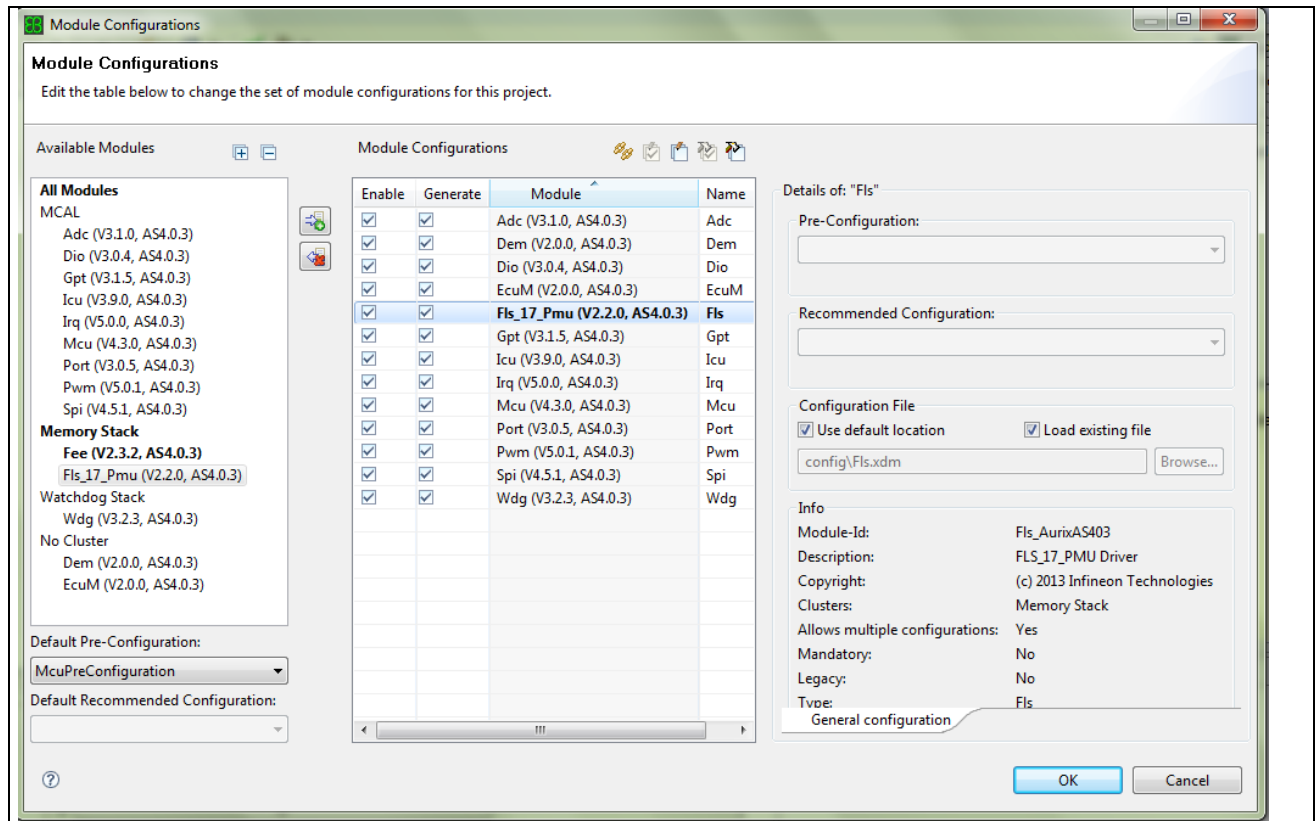


Figure 11 Adding FLS for demo application

Add the required modules, ensure that you click “Load existing xdm” to use the save configuration for demo application.

Similarly add the required modules. Ensure that you add a module only if you have installed the required driver. For example module like Fls will appear on the drop down due to the availability of configured xdm file, but the module can be generated only if MEM package is installed.

2.3.2.3 EcuM Configuration

This is not needed for PreCompile or Link Time Configuration modules.

For a module initialization like FLS,

- In EcuModuleConfigurationRef, add an entry to include the config root corresponding to the module.

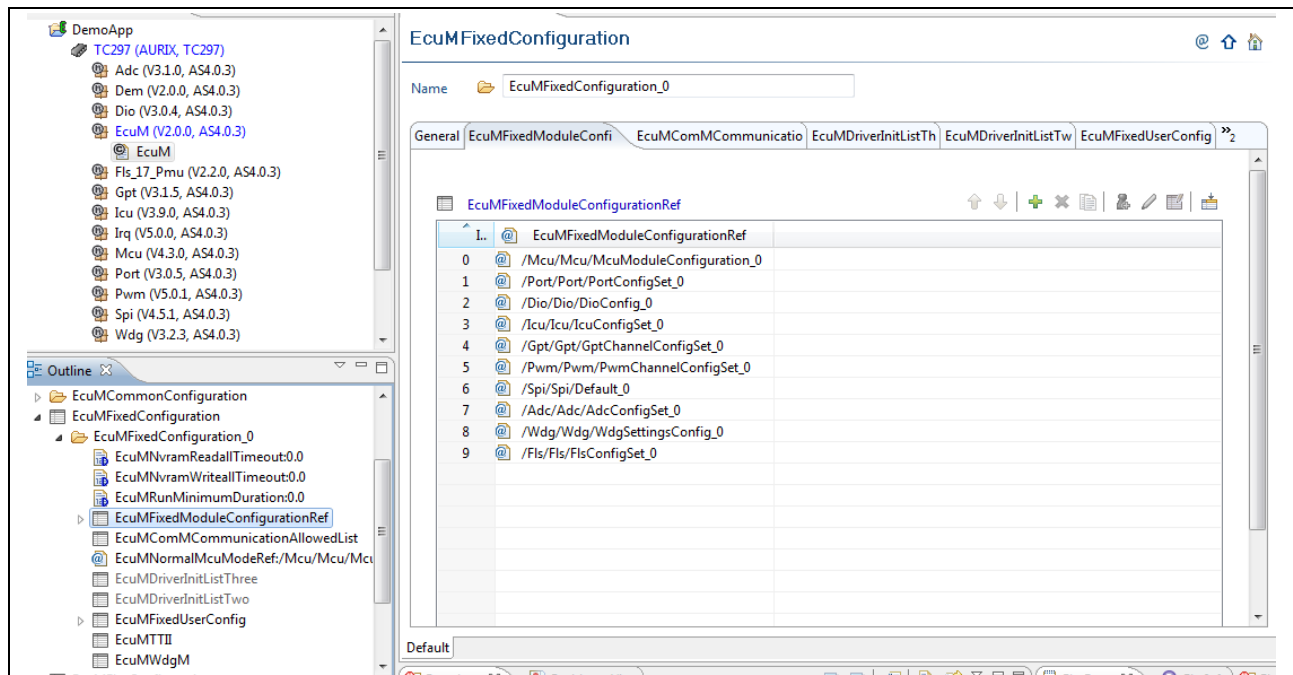


Figure 12 Adding the data to config root

- b) Goto EcuMDriverInitListOne, add a container EcuMDriverInitItem and give the required module ID and its reference.

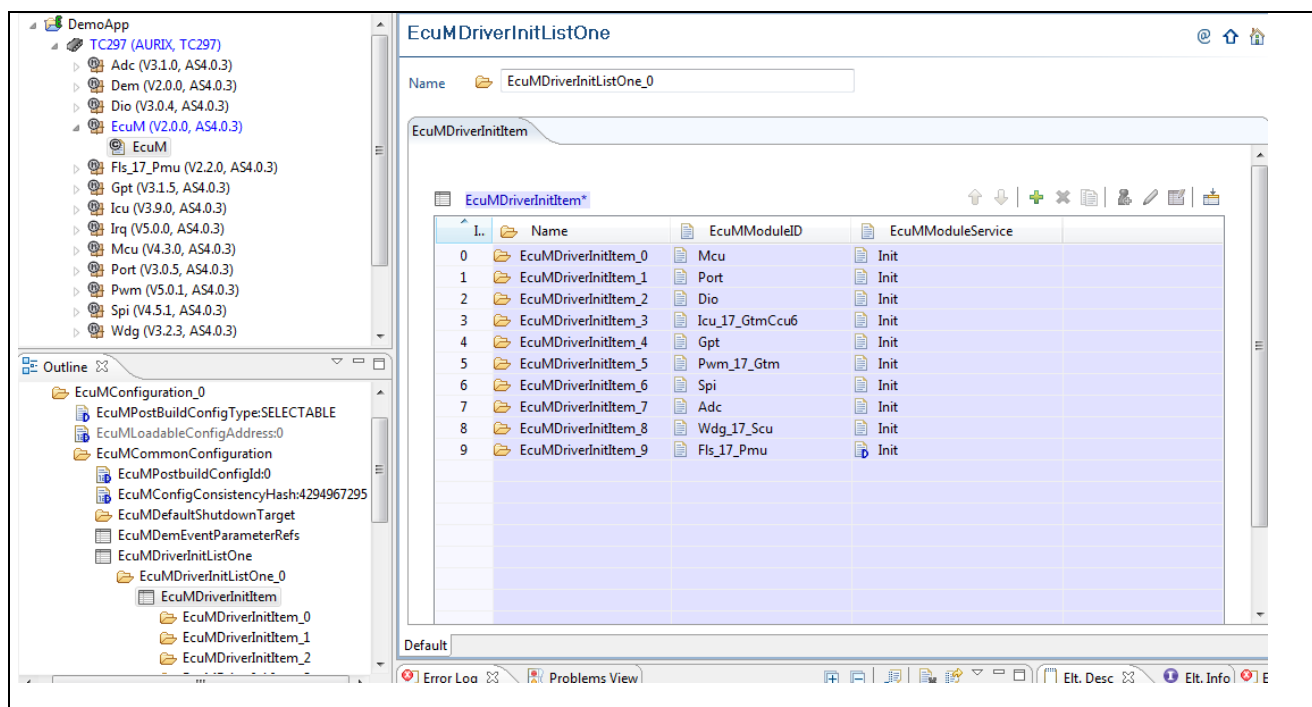


Figure 13 Adding Fls module in EcuM

Note: The name of the module entered in container "EcuMDriverInitItem" is case sensitive and should match the module name (driver name).e.g. Spi, Fls_17_Pmu etc.

To run DemoApp, Refer DemoApp.txt and ConfigurationReadMe.txt.

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